

Efficient Long-Video Understanding and Key Frame Selection: A Survey of Compression, Memory, and Reasoning

Kanchan Sarkar^{*1}, Darsh Agrawal², and Enakshi Jana³

¹Indian Institute of Technology

²Carnegie Mellon University

³Pondicherry University

Abstract

Long-video understanding requires locating sparse evidence, preserving temporal relationships, and reasoning over inputs that exceed practical multimodal context budgets. Increasing the number of input frames addresses only part of this problem: useful evidence may still be missed, compressed away, or incorrectly connected. This survey organizes representative work around five interacting mechanisms: query-aware frame selection, visual-token compression, persistent memory, hierarchical retrieval, and agentic evidence acquisition, with long-context scaling as a complementary systems axis. We introduce a common analytical formulation that separates candidate acquisition, retained visual tokens, memory, and reasoning cost. We compare architectural choices, summarize benchmark capabilities, and present source-reported numerical examples with their evaluation conditions rather than constructing an uncontrolled leaderboard. Particular attention is given to the distinction between temporal coverage and evidence concentration, and between reducing answer-model tokens and reducing end-to-end computation. We conclude with a reproducible evaluation protocol and research directions in uncertainty-aware acquisition, motion-preserving selection, multimodal evidence alignment, and verifiable reasoning. This is a literature synthesis and methodological perspective; no new model, experimental measurements, or benchmark results are claimed.

1 Introduction

A long video is a sequence of events rather than a large collection of independent images. A question may require a brief visual detail, a transition between two states, a sequence of actions, or evidence distributed across distant moments. Processing every frame is expensive, but choosing too few frames can destroy the evidence needed to answer. These two pressures make long-video understanding a joint problem of perception, evidence acquisition, representation, and reasoning.

Recent multimodal large language models (MLLMs) translate images or frames into visual tokens consumed by a language backbone. A practical video pipeline must decide which frames to decode, which visual features to encode, how many tokens to retain, and whether to revisit the source. These decisions occur at different computational stages. Frame selection operates before or around visual encoding; compression acts on representations or caches; memory retains information across time; retrieval exposes only relevant portions of a larger index; agents decide which additional evidence to request. Long-context models increase the capacity of the final reasoner but do not eliminate the need for these choices.

Earlier surveys cover video-language model families and the transition from image reasoning to extended temporal understanding [18, 29]. Our emphasis is narrower: how evidence is acquired and preserved under a finite computational budget. We examine interfaces between mechanisms, identify what reported efficiency measurements actually count, and distinguish benchmarks that test duration from tasks that require temporally distributed evidence.

Our synthesis makes three contributions. First, it provides a unified vocabulary for candidate frames, selected evidence, retained tokens, memory, and iterative acquisition. Second, it compares representative methods across mechanism and evaluation conditions. Third, it specifies a reporting protocol that separates selection quality, representation loss, and downstream reasoning errors. The protocol and design recommendations are analytical proposals, not empirically validated contributions.

2 Scope and Review Method

2.1 Scope

This survey concentrates on long-video question answering and related evidence retrieval with MLLMs. Summarization, temporal grounding, and streaming understanding are discussed where they illuminate the same architectural constraints. We consider representative 2023–2026 systems, with a literature cutoff of October 5, 2026. The review is a curated narrative survey, selected to expose distinct acqui-

^{*}Corresponding author: kanchansarkar9975@iitbombay.org.

sition, representation, and reasoning mechanisms rather than to enumerate every video-language model.

2.2 Source selection and extraction

We use primary papers and official proceedings to extract intervention points, query dependence, adaptation requirements, representations, benchmark settings, and reported results. Bibliographic metadata and release dates are checked against the corresponding records. Version-specific descriptions follow the version identified in the reference; in particular, ReUnit is described using its September 2026 revision. Numerical comparisons retain their original evaluation conditions. The later sections develop analytical comparisons and proposed diagnostics rather than new experimental results.

2.3 Interpretation boundaries

Mechanism descriptions summarize the cited work; failure modes and combined-system recommendations are our analysis. No shared-backbone reproduction is performed. Differences in model weights, frame count, resolution, subtitles, prompts, API snapshots, and splits prevent unrestricted comparison of published scores. Missing cost information is left unavailable rather than estimated from accuracy or context length.

3 Problem Formulation and Cost Model

3.1 Evidence under a budget

Let a video be $V = \{(x_i, t_i)\}_{i=1}^N$, where x_i is a frame or short clip and t_i its timestamp. Given query q , a system acquires a candidate set C , selects evidence $S \subseteq C$, builds representation Z , and produces answer $\hat{y}$:

$$C = A(V, q), \quad S = \pi(C, q), \quad Z = R(S, q), \quad \hat{y} = f(Z, q). \quad (1)$$

Here A may perform decoding or index lookup, π is a selection policy, and R may encode, compress, or summarize evidence. This abstraction covers a one-pass pipeline. Iterative systems additionally maintain state M_k and acquire new evidence conditioned on intermediate reasoning:

$$M_{k+1} = U(M_k, A_k(V, q, M_k)). \quad (2)$$

A generic design objective is

$$\max_{A, \pi, R, f} \mathbb{E}[\mathcal{U}(\hat{y}, y)] \quad \text{s.t.} \quad \mathbb{E}[\mathcal{C}] \leq B, \quad (3)$$

where utility $\mathcal{U}$ can reflect answer correctness, grounding, or calibration, and B is a budget. These equations are a survey abstraction, not a restatement of every cited method’s optimization problem.

3.2 What makes a video “long”?

Three quantities should be separated: source duration, representation length, and evidence span. A static hour-long recording can be easy to summarize; a three-minute manipulation sequence can require many state transitions. EgoSchema explicitly motivates temporal difficulty beyond clip duration [9]. A useful evaluation therefore reports duration together with the number and temporal distribution of evidence intervals required by each question.

3.3 Tokens are only one cost component

Let n_C be acquired candidates, n_S selected frames, and p_i the retained token count for frame i . The answerer’s sequence length is approximately

$$L = L_q + \sum_{i \in S} p_i + L_{\text{aux}} + L_M, \quad (4)$$

where auxiliary tokens may include subtitles, timestamps, or retrieved captions, and L_M is serialized memory. Counting frames alone cannot determine L when image resolution, cropping, or compression varies.

An end-to-end accounting is

$$\begin{aligned} \mathcal{C}_{\text{total}} = & \mathcal{C}_{\text{decode}} + \mathcal{C}_{\text{index}} + \mathcal{C}_{\text{select}} \\ & + \mathcal{C}_{\text{encode}} + \mathcal{C}_{\text{prefill}} + \mathcal{C}_{\text{generate}} + \mathcal{C}_{\text{tools}}. \end{aligned} \quad (5)$$

This decomposition prevents an answerer’s smaller context from being mistaken for a cheaper full system. A selector can encode hundreds of candidates even when the reasoner receives only eight frames. Caption-based retrieval may move computation into preprocessing rather than remove it.

For dense attention, attention arithmetic has quadratic dependence on sequence length in a simple model, whereas key-value storage scales approximately linearly in sequence length for fixed architecture. These are explanatory scaling relations, not predictions of wall-clock latency: optimized attention kernels, batching, encoder work, sequence parallelism, and memory bandwidth can change the measured bottleneck.

3.4 Illustrative arithmetic

At one sampled frame per second, a one-hour video yields 3,600 candidates. If each contributes 256 retained tokens, the visual input alone contains 921,600 tokens. A 64-frame representation at the same token density contains 16,384. This illustrative calculation is not a measurement of any reviewed method. It also omits candidate scoring, decoding, text, and iterative calls, showing why final-token reduction and total-cost reduction require separate reports.

4 A Taxonomy of Efficient Architectures

Figure 1 separates mechanisms by the representation they alter. These families are composable: an agent can query a tree, a tree can store compressed features, and a memory model can admit evidence chosen by an adaptive selector.

4.1 Timeline of representative methods

Figure 2 organizes representative methods into four parallel research strands. Dates indicate initial preprint release, with conference-only entries marked by publication year. The distinction explains why a method can appear earlier in the timeline than its eventual conference citation. ReUnit first appeared under the name Q-Fold in June 2026 and was substantially revised and renamed in September 2026. The timeline records milestones rather than implying that one strand superseded another.

2023: compact history and frame representations. MovieChat introduced a dense-to-sparse memory route, while LLaMA-VID explored a compact context/content token representation [6, 14]. These approaches targeted representation size rather than query-specific exploration of a large candidate pool.

2024: several directions develop in parallel. MA-LMM explored online historical memory; VideoAgent and DrVideo developed iterative acquisition; VideoTree used a query-adaptive hierarchy; LongVA and LongVILA increased context capacity; and Video-XL and LongVU reduced representation cost [1, 4, 8, 12, 13, 20, 22, 25]. Frame-Voyager added a learned frame-combination selector supervised through an answer model’s prediction losses [24]. The common question shifted from simply fitting a video into a prompt to deciding which information should be acquired, retained, or revisited.

2025: explicit allocation and task-aware representations. MDP3 emphasized list-wise relevance, diversity, and order; AKS balanced query relevance with source coverage; Q-Frame coupled frame choice and resolution; and Video-XL-2 adapted key-value access to the task [10, 15, 17, 26]. These methods operate at different stages, so their efficiency claims require different denominators.

2026: uncertainty, presentation, and richer scoring. FOCUS explores uncertainty-aware acquisition, ReUnit jointly allocates relevance and presentation granularity, and DAFS derives frame evidence from query-conditioned attention [16, 21, 28]. These approaches intervene at different interfaces and therefore need different controls when testing their accuracy and cost.

4.2 Uniform sampling as an explicit baseline

Uniform sampling offers predictable cost and broad temporal coverage without query conditioning. Its weaknesses are structural: brief events can fall between samples, repeated scenes can consume slots, and spatial detail can be lost at low resolution. Nevertheless, it remains a necessary baseline. Adaptive systems should be compared with a tuned uniform sampler under the same frame or token budget, not only with a weak sparse default.

Shot-based sampling provides a related query-independent baseline. It can improve scene diversity but need not preserve repeated actions within a shot. Motion-guided sampling can reveal transitions yet overallocate to camera movement. These are general design alternatives; neither is claimed here as a newly evaluated method.

4.3 Query-aware frame selection

Image-text embedding models such as CLIP supply a common way to estimate query-frame relevance [11]. However, semantic similarity is a proxy for answer utility. A frame depicting the queried object can score highly while omitting the action, relation, or before-after comparison required by the question.

Frame-Voyager. Frame-Voyager learns to query frame combinations using supervision derived from the prediction losses of a pretrained video answerer [24]. It represents a training-based alternative to embedding-only ranking. The supervision links selection to downstream answer utility, but can depend on the answerer’s checkpoint and training distribution. Cross-answerer evaluation is consequently an important diagnostic.

AKS. Adaptive Keyframe Sampling formulates a trade-off between relevance to the prompt and coverage of the source [17]. It is a plug-in selector intended to improve the information supplied under a fixed downstream budget. Its main conceptual role in this taxonomy is to move beyond relevance-only ranking. A general limitation of candidate-based selection remains: an informative event absent from the candidate pool cannot be recovered by ranking that pool.

MDP3. This training-free approach explicitly considers query relevance, diversity of the selected list, and sequential structure. It combines determinantal selection with segment-wise allocation and dynamic programming [15]. The contrast with independent top- k ranking is useful: the value of a frame depends on what has already been selected. In application, the usefulness of diversity depends on the feature metric and how well it separates genuine evidence from merely different backgrounds.

Q-Frame. Q-Frame couples query-conditioned selection with multi-resolution allocation [26]. Matching scores guide

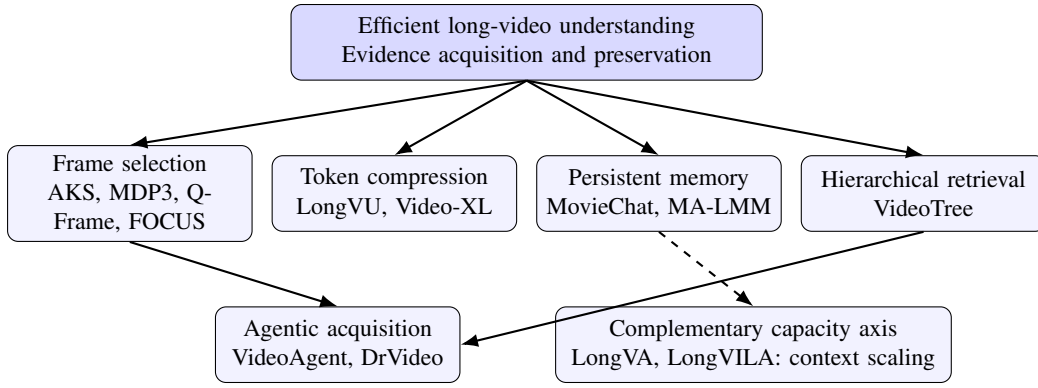


Figure 1: Survey taxonomy organized by the locus of intervention. The branches are complementary design choices rather than a chronological succession. Examples are drawn from sources cited in Sections 4.3–4.8. The diagram is an original synthesis.

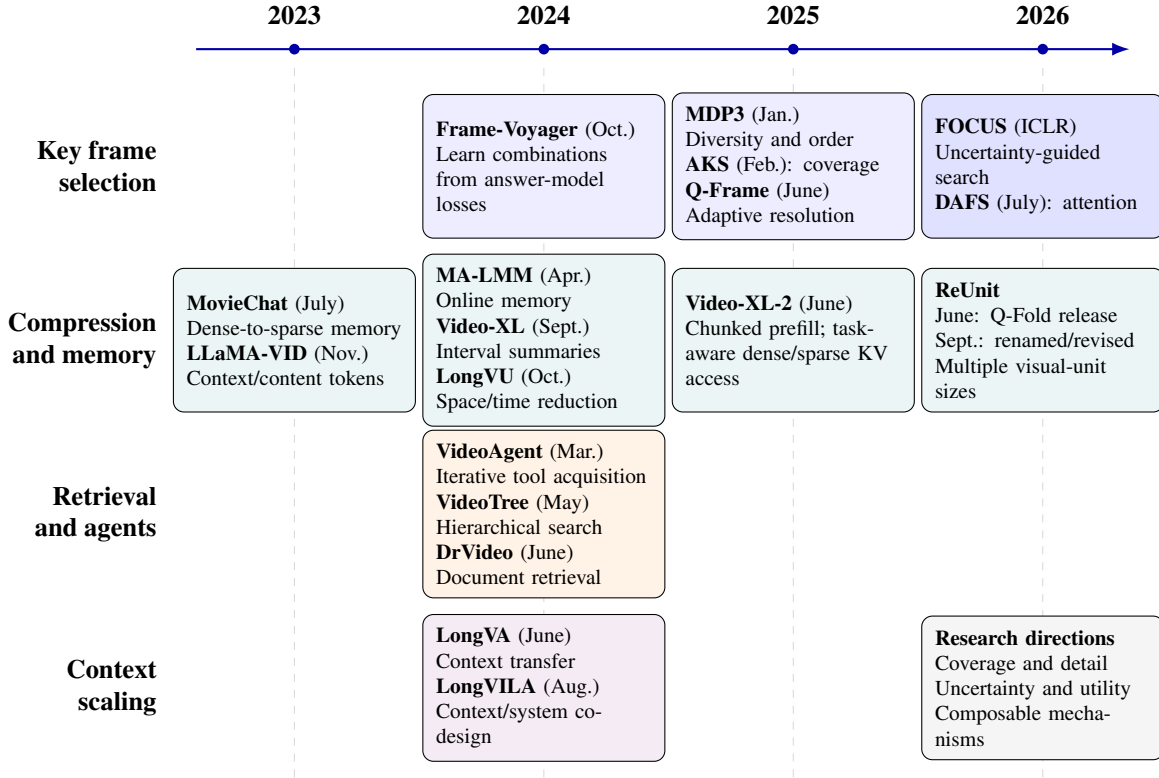


Figure 2: Timeline of representative methods, annotated with intervention mechanisms and initial release months. FOCUS uses its ICLR 2026 publication year; ReUnit also shows its September revision. Selection, representation, retrieval, and context scaling evolve in parallel. Empty cells do not imply absence of work. References are given in Tables 2 and 1. Original survey synthesis.

stochastic frame selection, while supported backbones allocate different spatial budgets to retained content. This makes temporal and spatial allocation jointly relevant. The paper distinguishes fixed-frame experiments from fixed-token experiments; proprietary GPT-4o examples use frame selection without its multi-resolution component. This distinction matters when interpreting the numerical examples below.

FOCUS. FOCUS treats short temporal regions as arms in a combinatorial pure-exploration formulation. Empirical

scores and Bernstein confidence radii guide exploration of uncertain regions before frames are selected within promising regions [28]. It represents adaptive exploration rather than only reranking a static candidate set. Its acquisition overhead and confidence calibration should be measured alongside downstream answer quality.

DAFS. DAFS extracts query-conditioned relevance from selected MLLM attention layers without autoregressive selection-time generation. It jointly allocates candidate

Table 1: Additional milestones included in the expanded timeline. Dates refer to initial arXiv release, with revisions identified when relevant. Contributions summarize primary sources; diagnostic questions are this survey’s analysis.

Method	Release	Main contribution	Evaluation question to test
LLaMA-VID [6]	Nov. 2023	Query-conditioned context and visual content tokens	Which local details survive a compact per-frame representation?
Frame-Voyager [24]	Oct. 2024	Learned frame combinations ranked using answer-model losses	Does the selector transfer to different answerer checkpoints?
Video-XL-2 [10]	June 2025	Chunked prefill and task-aware dense/sparse KV access	What is the full latency including KV reload and storage access?
ReUnit [16]	June/Sept. 2026	Query-aware visual units with several presentation granularities	Does compact context retain detail under an equal token budget?
DAFS [21]	July 2026	MLLM attention scoring and joint candidate/token allocation	Is selection useful after counting the selector’s own inference cost?

count and per-frame tokens using dynamic programming [21]. Attention-based evidence scoring differs from an independent embedding matcher, but attention weights do not by themselves establish causal evidence or answer correctness. The selector’s inference work belongs in full-system cost accounting.

4.4 Selection failure modes

Three tensions recur. First, concentration on a high-scoring region can omit evidence elsewhere. Second, visual redundancy and evidential redundancy are different: two similar frames may reveal a consequential state change. Third, an image selector can miss motion evidence that is apparent only in a short clip. Selection policies should preserve timestamps and, for action questions, permit local neighborhoods or clip-level evidence. These recommendations follow from the acquisition problem; they are not reported ablations of the reviewed methods.

4.5 Token compression and adaptive representations

Compression retains more source coverage while reducing the representation seen by the reasoner. The compression site matters: dropping frames removes temporal samples; spatial pooling removes local detail; learned summary tokens alter feature capacity; cache sparsification changes the context available to later computation.

LLaMA-VID. LLaMA-VID uses context and content tokens to compactly represent each frame [6]. This early example illustrates how a small visual interface can admit a longer source sequence. The question for compression is which query-relevant details remain recoverable in that interface, rather than only how many frames fit.

LongVU. LongVU combines removal of redundant frames using DINOv2 features with text-guided feature reduction and spatial reduction exploiting temporal dependencies [12]. It therefore spans both temporal and spatial

compression. The essential architectural point is staged reduction, rather than assuming one global pooling rule is appropriate throughout a video. A trade-off to test is whether small objects or transient changes disappear when neighboring frames look globally similar.

Video-XL. Video-XL uses visual summarization tokens to condense interval information into their associated key-value representations. Its training includes progressive compression, with interval granularity adapting the representation [13]. A compression ratio describes representation size in a particular configuration rather than a proportional latency improvement.

Video-XL-2 and ReUnit. Video-XL-2 separates chunk-based prefill from task-aware dense or sparse KV access during decoding [10]. ReUnit jointly chooses content and its presentation granularity, using visual units containing one, four, or nine source frames, each rendered as a standard image [16]. These approaches expose complementary choices: changing internal cache access versus changing the visual input interface. Input-unit counts and retained source-frame counts are consequently different quantities.

4.6 Long-context scaling

LongVA investigates transfer of extended language context to visual inputs, supporting a reported 2,000-frame or over-200,000-visual-token configuration [25]. LongVILA combines context extension, long-video supervised tuning, and multimodal sequence parallelism [1]. These approaches increase capacity rather than rely solely on removing evidence. Their significance is architectural and systems-oriented: training and serving must sustain long multimodal sequences. Capacity tests based on a planted visual needle are informative for retrieval but do not establish causal or multi-event reasoning ability.

4.7 Persistent memory

Memory systems update a bounded state as video content arrives. MovieChat uses a memory-inspired dense-to-sparse

representation [14]. MA-LMM processes content online and retains historical information in a memory bank that can be integrated with multimodal language models [4]. These designs offer a route to histories that cannot be represented as one monolithic prompt.

Our analysis distinguishes capacity from recoverability. A compact state can retain a scene’s topic while losing exact text, counts, or an object’s identity. Later queries may require details that appeared unimportant during ingestion. Retaining pointers to original timestamps permits a system to re-open evidence when the memory is insufficient. Compression without such access makes early mistakes irreversible.

For streaming tasks, evaluation should prohibit future evidence when answering a question at time t . Offline full-video methods and online memory systems have different information access. They should not be labeled equivalent solely because both maintain short prompts.

4.8 Hierarchical retrieval and agentic reasoning

VideoTree constructs a query-adaptive hierarchy and refines evidence at multiple granularities [22]. It exposes a useful middle ground between uniform global coverage and flat frame ranking: coarse summaries guide which regions deserve finer inspection. Its reliability depends on whether relevant details survive the coarse representation; a mistaken pruning decision can conceal an entire branch.

VideoAgent uses an LLM to plan iterative acquisition with vision–language tools [20]. DrVideo converts video into a coarse text document, then retrieves and augments relevant information through an iterative loop [8]. Both move part of the problem from context packing to deciding what to inspect next. A low final frame count can coexist with substantial captioning, indexing, or tool-call cost.

Caption intermediates improve accessibility to text retrieval but create a perceptual bottleneck. An omitted visual relation cannot be recovered by text-only reasoning over an incomplete caption. Agents should be evaluated for evidence sufficiency, revisitation behavior, and stopping decisions in addition to final answers. An agent that reasons fluently over unsupported captions may be confidently incorrect.

5 Key Frame Selection as Evidence-Set Construction

The selector families above can be compared through the evidence sets they construct. A useful comparison asks which observations are considered together, what proxy determines their value, and whether a rejected observation can be recovered. These questions cut across method names: learned selection, embedding matching, diversity optimization, attention scoring, and adaptive exploration can each be useful under different evidence requirements. This section develops a

common analytical vocabulary; the objectives below are explanatory models rather than implementations attributed to individual papers.

5.1 Relevance is necessary but incomplete

Let $r_i(q)$ be a relevance score for candidate i . Independent ranking chooses the highest-scoring candidates under a budget, implicitly treating their contributions as additive. This is reasonable when one frame contains a self-contained answer, but it fails when observations are complementary. For a question about an object’s destination, an earlier frame may identify the object while a later frame shows where it was placed. Neither observation can be evaluated solely by similarity to the wording of the question.

A generic evidence-set utility can therefore include both individual and joint terms:

$$U(S, q) = \sum_{i \in S} r_i(q) + \sum_{\{i, j\} \subseteq S} c_{ij}(q) - D(S). \quad (6)$$

Here c_{ij} represents complementarity and D penalizes redundant use of a limited budget. The signs and interpretation of these terms depend on the task. Near-identical images may be redundant for scene classification but complementary for detecting a subtle state change. An operational selector need not compute this equation directly; the formulation explains why average relevance can improve while answer quality does not.

Question decomposition provides one route to approximating complementarity. A query can imply a reference state, a transition, and an outcome. A policy can reserve evidence for each part before optimizing within a part. The difficulty is that linguistic decomposition can itself be wrong. For example, a question phrased with “after” can require identifying an event boundary rather than observing two arbitrary timestamps. Decomposition should consequently be evaluated against evidence annotations, and uncertainty in the inferred structure should permit broad fallback coverage.

The distinction also matters for negative questions. Establishing that an event never occurred requires coverage of the relevant interval, not merely failure to retrieve a matching frame. A ranking system cannot equate low similarity with absence. The appropriate response may be a broader search, a qualified answer, or abstention. This requirement changes both the selection objective and the meaning of a confidence score.

5.2 Coverage has several meanings

Temporal coverage measures how well selected timestamps span the source. Semantic coverage measures whether represented scenes or topics are varied. Evidential coverage measures whether the set contains the observations required for a particular answer. These quantities can disagree. Uniform sampling has strong temporal spread but may miss a brief action. A scene-diverse sampler can select many backgrounds without capturing the decisive state transition. A

Table 2: Architectural comparison of representative methods. “Training-free” concerns the added selection/reasoning procedure and does not imply untrained encoders or free inference. This table compares mechanisms, not standardized measured cost.

Method	Main mechanism	Adaptation	Representation/interface	Principal trade-off to examine
Frame-Voyager [24]	Learned frame combinations	Selector training	Loss-supervised frame list	Cross-answerer transfer
AKS [17]	Relevance and coverage	Training-free selector	Selected frames for an MLLM	Evidence absent from candidates
MDP3 [15]	List-wise diversity and order	Training-free selector	Segment-conditioned frame list	Feature diversity versus evidence utility
Q-Frame [26]	Query selection; resolution	Training-free selector	Mixed spatial budgets where supported	Backbone-dependent token accounting
FOCUS [28]	Uncertainty-aware exploration	Training-free selector	Region acquisition then keyframes	Acquisition overhead and calibration
LLaMA-VID [6]	Compact dual tokens	Model integration	Context/content tokens	Recoverability of fine details
LongVU [12]	Spatiotemporal reduction	Model/pipeline integration	Reduced visual features	Fine detail lost during reduction
Video-XL [13]	Learned summarization; KV	Compression training	Interval summary representations	Information recovery after compression
Video-XL-2 [10]	Task-aware KV access	Model/pipeline integration	Chunked prefill; dense/sparse KV	Cache reload cost
ReUnit [16]	Multi-granularity units	Training-free input design	One/four/nine-frame image units	Spatial detail versus temporal coverage
DAFS [21]	Attention-based selection	Training-free selector	Budgeted candidates and tokens	Selector inference overhead
LongVA [25]	Language context extension	Context/visual alignment	Long visual token sequence	Prefill and cache cost
LongVILA [1]	Context and system co-design	Long-video training	Distributed long context	Resource availability and serving scale
MovieChat [14]	Dense-to-sparse memory	Memory architecture	Short/long-term state	Exact detail versus compact state
MA-LMM [4]	Online memory bank	Memory integration	Historical features	Retention across changing queries
VideoTree [22]	Adaptive hierarchical search	Training-free framework	Tree-structured evidence	Early branch pruning
VideoAgent [20]	Iterative tool acquisition	Agent orchestration	Retrieved frames and tool outputs	Calls, latency, and stopping
DrVideo [8]	Document retrieval; refinement	Agent orchestration	Caption document plus augmented evidence	Caption omissions and indexing cost

highly concentrated selector can answer a local question well while failing a summary request.

For an annotated evidence set $E = \{e_1, \dots, e_m\}$, a diagnostic completeness score can be defined as

$$\text{Comp}(S, E) = \frac{1}{m} \sum_{j=1}^m \mathbf{1}[S \text{ adequately observes } e_j]. \quad (7)$$

“Adequately observes” must be specified by the annotation protocol: a timestamp overlap can be enough for a static scene but insufficient for legible text or an action requiring several successive observations. An all-evidence indicator complements this average by requiring every indispensable component. Reporting both prevents a selector from receiving a high average score while consistently omitting one critical part of multi-event questions.

Coverage should also be conditioned on the candidate pool. If annotated evidence is missing from the pool,

no reranking policy can recover it. Candidate recall and retained-evidence recall should therefore be reported separately. This distinction is central when comparing a static sampler with an acquisition policy that can reopen source intervals. A method with a richer candidate pool may provide a better system without being a better ranker on identical candidates.

Duration bins alone do not reveal this issue. A short interval hidden in a long static video has a different selection challenge from a long sequence of interacting events. An informative benchmark stratification combines source duration, evidence duration, evidence count, temporal separation, and required spatial precision. Such a stratification makes it possible to identify the mechanism responsible for an improvement rather than attributing every gain to “long-video understanding.”

5.3 Diversity, repetition, and identity

Diversity-aware selection reduces repeated allocation to visually similar candidates. MDP3 makes list-wise diversity and order explicit, whereas AKS balances relevance and coverage [15, 17]. The general benefit is a more varied evidence set under a finite budget. However, variation in an embedding space is a proxy, and its interpretation depends on what that space preserves.

Repeated actions are a useful stress case. Counting how many times a door opens requires retaining several similar transitions. A diversity objective may suppress these as duplicates, even though their multiplicity is the answer. Conversely, shots of different rooms can be visually diverse while carrying no information about the count. The selector should distinguish repetition that is evidentially meaningful from repetition that merely consumes context.

Identity-sensitive questions pose a related problem. If two visually similar objects exchange positions, a representation that preserves only category labels cannot resolve which object moved. Candidate diversity should not destroy links between observations of the same entity. Retaining timestamps, local trajectories, and relational context can help a reasoner connect these observations, but the memory or selector must also avoid inventing continuity across cuts.

An ablation can separate these effects by evaluating three diagnostic categories: duplicated views of an unchanged state, repeated instances of the same action, and transitions involving visually similar entities. A policy that improves only the first category provides redundancy reduction; success on the latter categories requires stronger temporal or relational preservation. This distinction prevents a broad diversity claim from being interpreted as a guarantee about temporal reasoning.

5.4 Order is a property of the evidence interface

Sorting selected frames by timestamp is a necessary interface choice for many temporal questions. It does not guarantee that the selection retained the transitions needed to infer order. If the selected set contains the first and last states but omits the action connecting them, the reasoner may fill the gap with a plausible narrative. Timestamp order constrains the observations; it cannot supply missing observations.

Chronological packing should preserve the original time scale rather than suggest equal intervals between adjacent images. An interface can include absolute timestamps, elapsed gaps, and clip boundaries. This is especially important for a sequence assembled from multiple distant moments. A model otherwise has to infer whether two adjacent images are milliseconds apart or separated by half an hour.

Temporal evaluation should distinguish ordering observed events from inferring unobserved causal relations. A reversal test can reveal sensitivity to order, but reversing an entire video may also produce unnatural motion. A cleaner diagnostic can keep the visual content fixed and permute the pre-

sensation of timestamped event units, while separately testing genuine source sequences. The first tests interface use; the second tests perception and evidence acquisition. Neither alone establishes causal understanding.

Timestamp granularity is another confound. A frame selected near an event can be assigned a rounded time that appears to precede the reference event even when it does not. Evaluation should retain precise source times internally and use consistent units in rendered prompts. Presentation labels can be rounded for readability without altering the underlying trace used for grounding metrics.

5.5 Candidate acquisition and adaptive refinement

Candidate generation sets the ceiling for downstream selection. A coarse scan can be efficient, but short events can fall between samples. One remedy is hierarchical refinement: inspect broad intervals, then allocate denser local observations where the query or uncertainty suggests they are needed. VideoTree illustrates a hierarchy, and FOCUS illustrates uncertainty-guided exploration [22, 28]. These are complementary examples of reducing unnecessary source inspection, with distinct representations and decision rules.

A hierarchical policy should retain a recovery route for regions initially judged irrelevant. Coarse summaries can miss a tiny object, a brief subtitle, or a transition whose significance becomes clear only later. A hard early rejection can make an apparently efficient search systematically incomplete. Keeping a modest coverage reserve or revisiting branches after contradictory evidence is a design hypothesis that can be tested with missed-evidence annotations.

Candidate density should be treated as an adaptive resource, not a constant hidden in preprocessing. A system can increase density around a detected transition while leaving stable intervals sparse. The cost depends on whether features for the denser samples already exist, whether the decoder must seek backward, and whether the new observations change earlier summaries. A useful trace records both requested density and actual decoded timestamps, because codec behavior can make the nominal request differ from the frames returned.

Stopping must account for uncertainty about unobserved regions as well as confidence in observed evidence. A convincing answer based on one scene does not demonstrate that another scene would not contradict it. Conversely, repeated inspection of an already sufficient interval can waste the budget. Policies should be tested on questions with absent evidence and on misleading initial observations, not only on examples where an answer is quickly found.

5.6 Spatial allocation and frame-equivalent budgets

The selected-frame count is an incomplete resource measure when frames have different resolutions or are combined into composite inputs. Q-Frame varies resolution, and ReUnit

varies presentation granularity [16, 26]. These choices expose a common trade-off: more source coverage can be delivered through a compact visual interface, but each observation may contain less readable spatial information.

For a retained token budget B_v , an illustrative allocation problem is

$$\max_{S, \{p_i\}} U(S, \{p_i\}, q) \quad \text{s.t.} \quad \sum_{i \in S} p_i \leq B_v. \quad (8)$$

The utility now depends on how much detail each observation retains. A label-reading question can have a sharp threshold below which additional temporal samples are useless. A route or activity question may benefit from many lower-detail observations. This means that a fixed-frame study and a fixed-token study answer different questions even with the same answerer.

The image processor can make token allocation discontinuous. Resizing, tiling, padding, and aspect-ratio rules can cause two images with similar pixel counts to produce different sequence lengths. Comparisons should count tokens after the actual processor and report source frames per input image when composites are used. A claimed “eight-image” budget is not automatically an eight-source-frame budget.

Spatial ablations should include questions about small text, object attributes, scene layout, and motion. A mixed-resolution policy can appear favorable overall while sacrificing a particular category. Reporting those categories reveals whether the policy trades away details that are important in a deployment. It also separates the benefit of selection from the benefit of simply using more source observations.

6 Compression and Memory: Preserving Recoverable Information

Selection removes observations; compression changes the observations retained. Memory adds a temporal commitment: information discarded during ingestion may be needed by a query that has not yet arrived. Comparing these choices requires asking what can be recovered, for which questions, and at what additional source-access cost.

6.1 The timing of the query changes the objective

Query-conditioned compression can devote capacity to the evidence required by a known question. Query-independent ingestion must preserve information for a distribution of future questions. A compact representation optimized for one question should therefore not be assumed suitable for an archive supporting arbitrary later queries. This distinction separates online ingestion from offline task-specific context packing.

For a query distribution $P(q)$, an explanatory retention objective is

$$R^* = \arg \max_R \mathbb{E}_{q \sim P(q)} [U(f(R(V), q))] \quad \text{s.t.} \quad |R(V)| \leq B_M. \quad (9)$$

The expectation makes the dependence on future workloads explicit. If training emphasizes broad summaries, precise counting or OCR questions may be poorly supported. A change in query distribution can consequently invalidate the memory design even if the video domain remains unchanged. Evaluation should vary both the visual distribution and the question distribution.

A hybrid archive can store a compact query-independent index while obtaining detailed, query-specific evidence later. This reduces the requirement that every possible answer be encoded in the initial state. It introduces other dependencies: source availability, retrieval reliability, decoding latency, and permissions to retain the original video. Those dependencies belong in the description of the system rather than being treated as implementation details unrelated to the method.

6.2 Where compression occurs

Compression before visual encoding reduces encoder work but may remove information that a stronger encoder would have recognized. Compression after encoding can exploit richer features but does not save the cost of producing them. Compression inside a language-model cache changes the representations available during reasoning and may have different reuse behavior across questions. These interventions should be compared at their actual locations in the pipeline.

Spatial pooling can suppress small objects and text even when global scene semantics are preserved. Temporal pooling can blur action boundaries or merge different instances. Learned summary tokens can preserve task-relevant information from their training distribution while offering little guarantee about rare future details. Cache reduction can retain useful global information but make exact source attribution difficult unless timestamps and mappings are carried through the reduction.

Recoverability should be tested by controlled queries whose required information is known. Hold the acquired source evidence fixed and vary only the representation. If the uncompressed answerer succeeds but the compressed version fails, the test isolates a representation-related failure more clearly than an overall leaderboard. It still does not prove that the information is absent from the compressed features: a different decoder might recover it. The operational conclusion concerns the tested representation–answerer pair.

This distinction is important when assessing claims about information preservation. A smaller representation with similar average accuracy may lose a narrow but critical category. Conversely, a representation can discard redundant pixels while retaining all evidence needed for a particular workload. Evaluation should describe the workload and inspect category-level changes rather than using a single average as proof of universal preservation.

Table 3: Evidence requirements and controlled selection diagnostics. The examples and tests are proposed analytical cases, not benchmark measurements.

Requirement	What must survive	Misleading proxy	Controlled test
Local detail	One legible object or text region	Global image similarity	Hold timestamps fixed; vary resolution
State change	Reference and outcome states	One highly relevant frame	Annotate both states and transition
Event order	Several identifiable events with times	Sorted but incomplete frames	Check evidence completeness and order separately
Repeated action	Distinct instances and boundaries	Feature diversity alone	Separate unchanged duplicates from repeated events
Identity tracking	Links between the same entity	Category labels alone	Use similar objects with different trajectories
Absence claim	Coverage of the applicable interval	No retrieved match	Include absent-event queries and abstention

6.3 Memory state and provenance

MovieChat and MA-LMM provide representative routes to bounded historical state [4, 14]. Their shared motivation highlights a broader systems question: whether a memory stores only a summary of what happened, or also enough provenance to recover the underlying observation. A summary such as “the person prepared food” is compact but cannot answer which utensil was used or how many ingredients were added.

A provenance-bearing memory can separate observations from interpretations. An observation records the source interval, modality, entity or event, and confidence. An interpretation links several observations to a tentative explanation. Keeping these layers distinct permits corrections when later evidence changes the explanation without pretending that earlier source evidence changed.

The memory also needs an explicit update policy. New observations may corroborate a previous state, supersede it, or contradict it. Simply appending every caption creates redundancy and may exceed the budget. Replacing an old summary can erase a state required for before–after reasoning. A useful policy retains transitions and source pointers even when stable states are consolidated.

Provenance has a cost. Timestamp maps, entity references, and source identifiers consume storage and may add prompt tokens. That overhead should be reported rather than assuming it is negligible. Its value can be tested through recovery success, correction latency, and the fraction of answers grounded in reopened source evidence. Such tests make memory quality observable beyond the final answer score.

6.4 Temporal consolidation without false continuity

Adjacent observations are often similar, which makes consolidation attractive. However, similarity across a cut does not establish that an entity persisted continuously. A memory can inadvertently connect two different people, locations, or object instances. This error can be especially diffi-

cult to detect after the original features have been discarded.

Consolidation should therefore preserve boundary uncertainty. A scene cut, occlusion, or camera change can trigger a tentative identity link rather than an irreversible merge. For tasks requiring identity, the system can request a local clip or retain extra features around the boundary. For a broad summary, less detail may be adequate. The appropriate policy depends on the expected questions and the cost of later recovery.

Repeated consolidation introduces another concern: errors can accumulate across generations of summaries. A summary of summaries can gradually replace observed detail with familiar narrative structure. A diagnostic can compare direct compression of the source with repeated incremental consolidation under the same final budget. Differences reveal sensitivity to the update path rather than just the size of the final state.

Memory evaluation should also include delayed queries. Ask about an early detail after progressively longer intervening histories, with similar distractors inserted later. This distinguishes simple capacity limits from recency bias and interference. If retrieval is allowed, record whether the system answered from memory or recovered the source, since these are different operating behaviors with different latency.

6.5 Text, features, and structured events

Text summaries are easy for language models to read and for lexical systems to index. They are also selective interpretations: a captioner may omit an unusual relation, miss an object, or fail to transcribe text. Feature memories retain information outside a fixed vocabulary but are harder to inspect. Structured event records expose entities, relations, and times, yet depend on extraction quality and a schema that may not anticipate later questions.

These representations can coexist. A text index can locate a rough interval, features can rank candidates within it, and a local clip can provide final evidence. The benefit should be evaluated through actual recovery behavior rather than inferred from the number of components. More components can also create inconsistent descriptions or make debugging

difficult when each stage reports its own confidence.

A structured memory should distinguish unknown values from absent values. If an object attribute was never observed, storing “not present” introduces a false negative. Similarly, an event whose boundary is uncertain should retain an interval or confidence range rather than a fabricated exact timestamp. These choices influence both retrieval and later reasoning because downstream models often treat serialized records as established facts.

The interface should preserve the distinction between source text and generated text. A transcript may be erroneous but is derived from a recording; a caption is a model’s interpretation of visual evidence. Mixing them without labels can allow an interpretation to appear as a quoted source statement. Modality and provenance tags provide the reasoner with information needed to assess conflicts.

6.6 Budget allocation over memory lifetime

A fixed memory size does not imply fixed cost. Frequent updates, feature recomputation, consolidation calls, and source revisitation can dominate a small stored state. A complete report should disclose update frequency, work per update, retained storage, and source-access rate. Offline ingestion and real-time ingestion have different constraints even if they produce similar final representations.

Memory policies can be compared through retention curves: performance on a fixed early event as intervening content grows. A separate recovery curve measures success and latency when the source remains available. The first tests the state itself; the second tests the complete memory–retrieval system. Their separation prevents source access from concealing poor intrinsic retention while still crediting a useful deployment design.

The lifetime of a representation matters when models are upgraded. A feature index built with one encoder may not be directly compatible with a new query encoder. Generated captions can remain readable but preserve the old captioner’s omissions. Rebuilding the index incurs a potentially substantial cost. Systems should identify the model version associated with stored representations and measure the effect of upgrades on retrieval consistency.

Memory design thus involves more than compressing history. It determines which claims remain inspectable, how errors are corrected, and what future workloads can be served. A useful survey comparison should describe those commitments alongside architecture and final accuracy.

7 Systems Constraints Behind Apparent Efficiency

The cost model in Section 3 separates computation stages, but real implementations also depend on data movement and scheduling. A selector can reduce answer-model work while shifting the bottleneck to decoding, feature construction, storage, or repeated model calls. This section connects

architectural choices to measurable serving behavior without assigning unreported hardware costs to individual methods.

7.1 Video decoding and random access

Source-frame access is not free. A request for a frame at an arbitrary timestamp can require decoding preceding frames, depending on the codec and keyframe layout. A policy that repeatedly jumps between distant intervals may therefore incur more work than a sequential pass returning a similar number of observations. Counting only frames retained by the answerer misses this difference.

Decode evaluation should record the source format, resolution, frame rate, decoder, and access pattern. A pre-extracted image archive changes the problem: random access becomes simpler, but storage and preprocessing costs move outside the query path. Both settings can be useful, provided the baseline has the same source access. Comparing a cached adaptive method with an uncached uniform method would confound algorithm choice with data preparation.

Timestamp fidelity also matters. Variable-frame-rate inputs and approximate seeking can return observations different from the intended candidates. The trace should record the actual decoded times and preserve their association with prompts and evidence labels. Otherwise, a method may appear to miss an annotation because of a decoding mismatch rather than a selection failure.

An efficient acquisition schedule can group nearby requests and reuse decoded neighborhoods. This is a systems optimization whose contribution should be isolated from the selector’s accuracy. A matched policy evaluated with sequential and random-access decoding can reveal how much latency comes from the access pattern. Such a test is particularly relevant for agents that revisit intervals after each reasoning round.

7.2 Encoder work and reusable features

An embedding selector often encodes more candidates than the answerer receives. Those candidate features may be reusable across questions, but query-conditioned visual features may not be. The distinction determines whether an archive workload benefits from amortization. A complete method description should identify what is computed once, what is recomputed per query, and what can be shared with the final visual encoder.

Two encoders can also impose incompatible image processors. A low-resolution scoring feature may rank a candidate correctly, while the high-resolution answerer sees a different crop or aspect ratio. A claimed benefit from selection can then depend on preprocessing alignment. Recording crop rules and selected source regions helps isolate this interaction.

Feature reuse should not be assumed merely because both stages use a vision backbone. The projection layers,

precision, tokenization, and query conditioning can differ. Reusing a feature computed for one purpose may require re-training or reduce accuracy. Conversely, independently encoding identical frames can create avoidable overhead. An implementation audit should report actual reuse rather than an architectural opportunity for reuse.

Persistent features also introduce storage and transfer costs. A large index can exceed accelerator memory and require host-memory or remote-storage access. Retrieval latency then depends on locality and concurrent workloads. Accuracy measured with an index already resident on the accelerator may not predict latency for a larger archive. Evaluations should specify index placement and include transfer work in the relevant deployment setting.

7.3 Attention arithmetic and memory traffic

Long visual sequences stress both computation and memory movement. FlashAttention provides an example of exact attention designed around input/output efficiency, reducing transfers between GPU memory levels through tiling [2]. This illustrates why a theoretical attention operation count alone does not determine runtime. A change in kernel implementation can alter latency without changing the evidence presented to the model.

Sequence compression and efficient attention consequently address different constraints. Compression changes the information available to the reasoner; an exact attention implementation changes how the same attention computation is executed. They can be combined, but a study should separate their effects. If a compressed system uses a different kernel or precision from its baseline, the measured speed difference cannot be attributed entirely to information reduction.

The key-value cache introduces another resource. For a fixed architecture, a rough cache-size model is

$$M_{KV} \approx 2n_\ell L h_{KV} d_h b, \quad (10)$$

where n_ℓ is the layer count, h_{KV} the number of key/value heads, d_h the head dimension, and b bytes per stored element. The factor two represents keys and values. This formula excludes model weights, visual features, activations, allocator overhead, and architecture-specific cache formats. It is explanatory accounting rather than a peak-memory prediction.

PagedAttention addresses cache management and sharing in language-model serving [5]. It motivates separating representation sparsification from allocation efficiency: reducing unused allocation does not necessarily discard tokens, whereas pruning cache content changes what later computation can access. Reports should state whether a memory improvement concerns allocated bytes, live cache elements, or total process memory.

7.4 Prefill, generation, and concurrent workloads

Visual inputs can make prefill expensive even when the final answer is short. An iterative agent can repeat prefill across rounds, while a long-context model may pay a larger cost once. Total latency depends on the amount of shared prefix reuse and whether visual features or caches are retained between rounds. Call count alone is therefore not sufficient to compare an agent with a single-pass model.

Measure source-to-answer latency alongside time to first token and output-generation time. For a system that builds an index, record ingestion latency separately and include it in cold-start results. If partial answers are streamed while evidence acquisition continues, define whether the reported completion time refers to a provisional answer or the final verified answer. These operational definitions prevent incompatible latency claims.

Throughput under concurrency can differ from isolated latency. A method with a small per-query context may allow larger batches, while repeated tool calls can create scheduling overhead or dependency stalls. A service should evaluate both a single-query setting and a specified concurrent workload. Batch size, arrival process, and timeout policy should be reported because they affect tail latency and completed-query counts.

Adaptive policies also produce variable work. Easy questions can stop quickly; difficult questions may consume the full budget. Reporting only mean latency hides expensive tails. Percentiles and budget distributions show whether the system meets a deadline reliably. Failed or timed-out requests should remain in the accounting rather than disappearing from the reported accuracy set.

7.5 Distributed context and communication

Context scaling can distribute sequence processing across devices. LongVILA combines model development with sequence-parallel systems design, while the Large World Model work develops million-token video-language context with blockwise RingAttention [1, 7]. These examples show that input capacity depends on hardware organization and training as well as on a nominal context limit.

A larger supported sequence does not establish that every token is equally useful or that the deployment is cheaper. Distributed attention introduces communication, synchronization, and placement decisions. Its cost depends on the interconnect and how much useful computation overlaps communication. A report should name the device topology and count all devices involved, rather than quoting latency without the associated resource footprint.

Capacity and evidence preservation should also be evaluated separately. A model may retrieve a planted visual needle from a long context but fail to link several natural events. Position-controlled retrieval tests expose context access; multi-event questions test integration. Neither result substitutes for the other. The long-context approach should

be compared with acquisition and compression at matched deployment resources where possible, while acknowledging that matching token count alone would erase its intended capacity advantage.

Distributed serving may be appropriate for a shared archive service and impractical for an individual upload. The survey taxonomy therefore describes capabilities without assigning a universal deployment ranking. The relevant choice depends on workload, hardware, response deadlines, and whether source evidence can be revisited.

7.6 Cost frontiers and break-even reuse

Suppose a retrieval system has build cost T_b and per-query cost T_r , while a direct system has per-query cost T_d . Under a simplified fixed-workload model, retrieval becomes faster in aggregate when

$$Q(T_d - T_r) > T_b. \quad (11)$$

If $T_d \leq T_r$, no positive reuse count satisfies this timing argument. If $T_d > T_r$, the break-even count is $T_b/(T_d - T_r)$. This is an analytical relation, not a measured crossover for any reviewed method. It omits index maintenance, source changes, parallel scheduling, and storage charges.

A meaningful operating frontier plots accuracy against complete-system cost at several budgets. A method can be preferable at a very small budget and lose its advantage as additional evidence becomes affordable. The choice of one operating point can conceal this behavior. Budget sweeps should include both small and generous regimes, and methods that cannot run within a regime should be marked infeasible rather than assigned extrapolated results.

Cost comparisons should include uncertainty when timing variability is material. Warm-up, cache state, and external service load can alter measurements. Multiple runs on a fixed workload, consistent synchronization, and clear inclusion rules make the results easier to interpret. These requirements are particularly relevant for API-based systems, whose serving implementation is outside the experimenter’s control.

8 Reasoning over Retrieved Evidence

Retrieval changes the reasoner’s task from interpreting a complete sequence to integrating a selected collection of observations. The resulting interface can include images, captions, event records, transcripts, and summaries at different temporal scales. These pieces need enough structure to support comparisons and enough provenance to prevent unsupported details from becoming established facts.

8.1 Coarse-to-fine retrieval and revisitation

A hierarchy offers broad coverage without placing every detail in the final prompt. Coarse representations can identify regions for inspection, but their omissions should not be

treated as evidence that a detail does not exist. The hierarchy needs a way to return to a rejected branch when the question changes or later evidence contradicts the initial interpretation.

Revisitation can operate at several levels. A system can retrieve another summary, inspect a finer feature representation, decode a local image neighborhood, or acquire a clip with audio. Each action has a distinct cost and evidential value. An agent should therefore choose an evidence operation rather than only choose a timestamp. Asking for a higher-resolution crop differs from asking for a longer temporal window.

The action interface should expose what information is already available. Repeatedly requesting the same interval can be useful if the modality or resolution changes, but wasteful if the request merely repeats an identical observation. Traces should distinguish these cases. A duplicate-call rate without this distinction would penalize legitimate refinement alongside inefficient loops.

An unsuccessful retrieval also needs an interpretation. A timeout, an empty index result, and a confidently negative observation are different events. If all are serialized as “not found,” the reasoner can incorrectly infer absence. Tools should return status and coverage information in addition to content, enabling the model to know whether an interval was actually inspected.

8.2 Evidence sufficiency and stopping

Stopping rules should test whether the evidence supports the requested relation. A local attribute question can be complete once a relevant region is legible. A sequence question may require all event units and their order. An absence question can require broad coverage. There is no single universal frame count at which evidence becomes sufficient.

The agent’s confidence in an answer can diverge from evidence sufficiency. Multiple-choice options can make a plausible answer easy to guess; a fluent summary can hide missing observations. A sufficiency test should therefore refer to the claims that must be supported, not only to the probability assigned to an answer. Evidence labels or controlled diagnostic cases make this distinction measurable.

For budget-limited systems, stopping is also a resource decision. A system may exhaust its budget before becoming certain. The output should preserve that uncertainty rather than convert budget exhaustion into a confident answer. Evaluation can report answer coverage, accuracy among answered questions, and resource use, while separately tracking cases that stop because of a deadline or unavailable source.

The value of additional acquisition depends on what is missing. If an object is visible but illegible, another low-resolution frame may add little. If event order is ambiguous, a local clip around the boundary may be valuable. A practical policy should distinguish these uncertainty types. Testing targeted refinement against indiscriminate additional

Table 4: Accounting boundaries for complete-system efficiency. Each row identifies an observable cost; none is an estimate of a reviewed method’s hardware use.

Stage	Work to include	Reuse condition	Common omission
Source access	Transfer, seek, decode, resize	Same source and retained decoded data	Random-access decode work
Index construction	Features, captions, storage writes	Multiple queries with a compatible index	Build time hidden as preprocessing
Selection	Query embedding, scoring, optimization	Some candidate features can be shared	Scored candidates not counted
Visual encoding	Actual selected images, crops, clips	Compatible retained encoder features	Duplicate encoding in two stages
Reasoning	Prefill, generation, cache transfer	Valid prefix or cache reuse	Repeated agent-round prefill
Recovery	Extra source calls, tools, retries	Depends on requested intervals	Failed or repeated acquisitions
Maintenance	Index updates, model migration	Stable source/model versions	Rebuilding after an encoder change

frames can reveal whether an agent actually uses its intermediate state to allocate resources.

8.3 Cross-modal consistency

A spoken statement can describe an event that is not visible, a subtitle can be delayed, and OCR can repeat a title that no longer matches the scene. Multimodal evidence should therefore carry modality-specific times and source status. Concatenation into a single unstructured text block removes information needed to identify these differences.

Conflict handling should distinguish contradictions from complementary perspectives. A narrator saying “we will open the package” before the package is opened is not a contradiction. A caption claiming that the package is already open can be an interpretation error. The reasoner needs temporal context to resolve the difference. Evaluations should include such asynchronous cases rather than testing only perfectly aligned modalities.

Source priority depends on the question. For what was said, audio or a verified transcript is central. For what was visible, a narration is not a substitute for visual evidence. For a comparison between speech and action, both are required. A system can perform well by routing each question to an appropriate modality while still failing joint alignment; modality ablations and cross-modal questions should be reported separately.

Generated captions should not acquire authority merely through repeated use. If the same caption is retrieved in several rounds, it remains one model interpretation of one source interval. Repetition does not constitute independent confirmation. A trace can identify the original source and prevent repeated derived text from being counted as several corroborating observations.

8.4 Grounded outputs and error correction

An inspectable answer can associate each factual claim with a source interval. This supports auditing, but timestamp links must be evaluated for correctness. A model can answer correctly while attaching the wrong interval, or attach

an approximately relevant interval that lacks the decisive detail. Grounding quality is an additional metric rather than a formatting property.

Claim granularity should match the evidence. A broad summary sentence can refer to a long interval, while a precise attribute claim needs a tighter observation. Requiring a single timestamp for a complex multi-event sentence can force an artificial annotation. A better representation can associate one claim with several evidence intervals and indicate the role of each.

Correction behavior is equally important. If newly acquired evidence contradicts an earlier answer, the system should update the answer and its support links. Keeping obsolete evidence in the prompt without marking its interpretation as superseded can produce an internally inconsistent response. A diagnostic can reveal a second decisive observation after an initial provisional answer and measure whether the model corrects the relevant claim.

This reasoning interface makes the architecture easier to evaluate. Acquisition logs identify what was observed; representation mappings identify what survived; claim links identify what the answer used. The resulting analysis can credit a useful complete system while still locating its failures at the appropriate stage.

9 Benchmarks and Evaluation Coverage

9.1 Complementary benchmark questions

LongVideoBench focuses on interleaved video–language input and referred reasoning across extended context [23]. Video-MME spans diverse domains and durations and supports assessment with additional modalities [3]. MLVU broadens the task and video-genre mix [27]. LVBench stresses extreme-duration understanding and information extraction [19]. Together they reveal different failure modes; a single overall score cannot replace their task-specific analysis.

Table 5 summarizes verified scope information. We

separate source facts from recommended diagnostic uses. Benchmarks also have evolving splits and processing conventions. Reports should name the release, split, and modalities rather than referring to a dataset acronym alone.

9.2 Modality and temporal leakage

A subtitle-assisted answer can be correct without locating the corresponding visual evidence. Conversely, a visually grounded method may appear weaker on questions that depend on speech. Therefore visual-only, visual-plus-subtitles, and audiovisual settings should be reported separately. Subtitle timestamps, OCR, audio preprocessing, and transcript truncation can change both performance and cost.

For temporal grounding, answer correctness and localization correctness are distinct. A model may choose the correct option using a global semantic prior while pointing to the wrong interval. Timestamp supervision and evidence annotation make this distinction observable. Selection-quality metrics should be conditioned on the actual evidence requirement rather than every frame depicting a related object.

Table 5: Benchmark coverage. Counts and durations are source-reported; diagnostic uses in the final column are this survey’s recommendations. Counts refer to the cited benchmark versions.

Benchmark	Verified scope	Main emphasis	Recommended diagnostic use
EgoSchema [9]	Over 5,000 questions; three-minute clips	Egocentric multi-choice QA; intrinsic temporal difficulty	Distributed action evidence; context beyond a single moment
LongVideoBench [23]	6,678 questions; 1,337 validation and 5,341 test	Referred reasoning and interleaved subtitles	Cross-moment references; visual-only versus subtitle-assisted input
Video-MME [3]	900 videos; 2,700 questions; 11 s to 1 h	Domains, duration, video/audio/subtitles	Duration-stratified accuracy; modality ablations
MLVU [27]	Multiple task types and video genres	Broad long-video understanding	Task sensitivity to selection and compression
LVBench [19]	1,549 questions; average video length 4,101 s	Extreme-duration understanding	Sparse evidence search and long-horizon retention
V-NIAH [25]	Synthetic visual needle retrieval	Context access under distraction	Position-dependent retrieval; not a proxy for all reasoning

10 Published Performance and Efficiency: What Can Be Compared?

10.1 Within-source numerical examples

Table 6 lists reported examples rather than a unified leaderboard. Q-Frame permits useful within-paper comparisons because baseline and augmented versions share specified experimental conditions. VideoAgent and VideoTree use different frameworks and must not be ranked from these entries alone. The examples illustrate reporting requirements rather than exhaustively representing every method.

10.2 Efficiency claims need a denominator

Four ratios answer different questions:

$$\rho_{\text{frames}} = n_S/N, \quad \rho_{\text{tokens}} = L_{\text{visual}}/L_{\text{reference}}, \quad (12)$$

$$\rho_{\text{latency}} = T/T_{\text{reference}}, \quad \rho_{\text{cost}} = \mathcal{C}/\mathcal{C}_{\text{reference}}. \quad (13)$$

A method can improve one while worsening another. We therefore do not convert a selected-frame fraction, a representation compression ratio, or a model’s context capacity into a GPU-hour saving. Exact training GPU-hours and end-to-end dollar costs are not established consistently by the sources; this survey leaves unavailable values unreported.

10.3 Cold-start and amortized costs

For an index built once and reused for Q queries, a simplified per-query accounting is

$$\bar{T}(Q) = T_{\text{build}}/Q + T_{\text{retrieve}} + T_{\text{answer}}. \quad (14)$$

This relationship explains a deployment distinction, not a measured result: preprocessing-heavy retrieval can be attractive for frequently queried archives but unattractive for a one-off upload. Report Q , index lifetime, storage footprint, and invalidation policy whenever an amortized cost is claimed. A live stream also has an ingestion deadline that offline preprocessing does not face.

Table 6: Selected source-reported examples. No experiments were run for this survey. “pp” denotes an arithmetic difference between percentages. Results across separate source blocks are not directly comparable.

Source / setting	Benchmark	Reported result	Conditions and interpretation
Q-Frame: Qwen2-VL-7B [26]	MLVU	56.9 → 65.4	Fixed input-token comparison; +8.5 pp; resolution allocation changes retained frame count
Same setting [26]	LVB validation	53.5 → 58.4	No interleaved subtitles; +4.9 pp
Q-Frame: GPT-4o [26]	LVB validation	53.3 → 58.6	Eight input frames; frame selection only; +5.3 pp
VideoAgent [20]	EgoSchema / NExT-QA	54.1 / 71.3	Zero-shot; average 8.4 / 8.2 frames reported; tool cost requires separate accounting
VideoTree [22]	EgoSchema / NExT-QA	61.1 / 75.6	Test accuracy in the cited version; different pipeline from VideoAgent
LongVILA-7B [1]	Video-MME	65.1	Subtitle-assisted setting; not comparable to visual-only configurations

11 A Reproducible Evaluation Protocol

We propose three evaluation tracks that isolate complementary questions. They should be reported separately, because each permits different optimization freedoms.

11.1 Track A: selector isolation

Fix the answerer checkpoint, decoding settings, prompt, candidate pool, timestamps, image processing, and selected-frame budget. Compare uniform, relevance-only, and adaptive selection on exactly the same candidate set. This measures how effectively each selector uses available evidence. A method that changes candidate acquisition should additionally report Track C, rather than being credited only as a better reranker.

11.2 Track B: representation isolation

Fix source access and the answerer while varying spatial resolution, token pooling, learned summaries, or cache reduction under a shared final visual-token budget. Report actual tokens after the backbone’s preprocessing, not a nominal frame-equivalent count. Keep auxiliary text and subtitle budgets controlled. Include a matched-frame evaluation when possible to distinguish improvements from allocating more temporal samples.

11.3 Track C: complete-system evaluation

Permit adaptive decoding, indexing, agents, or memory. Measure the full pipeline from source availability to answer completion. Include cold-start and warm-index conditions, all tools and retries, and the effect of repeated queries. Compare accuracy–latency and accuracy–memory frontiers rather than one isolated operating point. For API systems, record service snapshot, input/output token charges, call count, and concurrency limits.

11.4 Reporting dimensions

Every track should disclose: dataset version and split; backbone and checkpoint; candidate count and sampling rate; selected frames or clips; retained tokens and resolution; subtitle/audio/OCR access; prompts and answer extraction; hardware and precision; batch size and cache conditions; random seeds; wall-clock latency distribution; and any index construction or model training cost.

For stochastic selectors, repeated runs should quantify variation. For paired evaluation on identical questions, paired bootstrap intervals are useful; when several questions share one source video, resampling at video level avoids treating dependent questions as fully independent. Publish per-question predictions and budget traces where dataset licenses permit.

11.5 Evidence-aware diagnostics

Partition errors into acquisition failure, representation failure, and reasoning failure. Acquisition failure occurs when necessary source intervals are never inspected. Representation failure occurs when acquired evidence is not retained with sufficient detail. Reasoning failure occurs when adequate evidence is available but the answer is incorrect. This attribution requires evidence annotations or a controlled diagnostic subset; it should not be inferred from attention maps alone.

Figure 3 shows how timestamped traces permit these distinctions. A useful trace records requested intervals, observed modalities, retained representations, compression decisions, tool outcomes, and stopping rationale. Trace logging itself adds overhead and should be included in production-cost measurements when enabled.

11.6 Controlled stress cases for mechanism attribution

Natural benchmarks establish broad task performance, but controlled cases can isolate why a system fails. The following scenarios specify diagnostic designs rather than measured outcomes. Each design should retain a natural-video

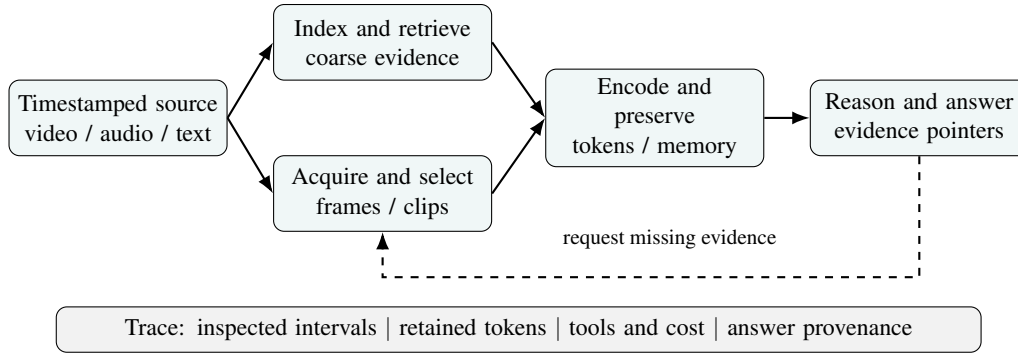


Figure 3: Original analytical diagram of a traceable long-video system. Retrieval and selection are alternative or combined acquisition paths. The dashed feedback permits evidence revisitation. Recording source timestamps at each interface helps distinguish missed evidence, lost representation detail, and faulty reasoning.

evaluation alongside the controlled variation, because artificial interventions can introduce distribution shifts of their own.

A brief event inside a stable scene. Choose a question whose evidence occurs in a short interval surrounded by visually similar content. Vary candidate sampling density while keeping the final selected budget fixed. Annotate whether the event enters the candidate pool and whether the selector retains it. This separates acquisition recall from ranking quality. A second variation keeps the event in every candidate pool and changes only the distractor count, testing whether scoring remains useful as the search space grows.

A detail that requires spatial precision. Use an interval containing a small object attribute or readable label. Keep the selected timestamps fixed while changing image resolution or token reduction. Record whether the answerer can read the detail from the uncompressed evidence before interpreting failure as compression loss. The same source can support a broad scene question as a control. If broad recognition remains intact while detail recognition declines, an overall average would conceal a task-specific cost of compression.

Two indispensable distant states. Select a question that requires identifying an earlier state and a later outcome. Annotate both intervals and test evidence-set completeness. Provide either state alone as a control to estimate whether the model can guess the answer from incomplete evidence. The complete-evidence condition should preserve timestamps and identity information. A selector that improves relevance but repeatedly keeps only the outcome can then be distinguished from one that preserves the relation the question actually asks about.

Repeated actions with similar appearances. Construct a counting or ordering diagnostic with several instances of the same action. Preserve the source event boundaries and

vary whether the candidate list also contains unchanged duplicate views. The first variation tests multiplicity preservation; the second tests redundancy suppression. Treating both as a single diversity benchmark would favor policies that discard repeated events. Report count error and instance coverage alongside answer accuracy, and inspect whether a model merges distinct events in memory.

A changed query after ingestion. Ingest a video without revealing the final question, then ask one broad query followed by a precise question about an earlier detail. Compare answering from stored memory with answering after permitted source recovery. Record additional observations and recovery latency. This exposes whether query-independent memory supports the future workload and whether revisitation repairs omissions. Reusing a representation built specifically for the final query would not test this property.

Evidence that overturns an initial interpretation. Present an ambiguous early observation, obtain a provisional answer, then reveal a later observation that resolves the ambiguity. Measure whether the answer and support intervals are updated consistently. A system that keeps its original answer despite decisive new evidence has a correction failure; a system that changes the answer without grounding it can still have a verification failure. The protocol should avoid scoring hidden reasoning text and instead assess observable claims, acquired evidence, and final support links.

Unavailable or genuinely absent evidence. Include separate cases for an absent event, a failed source-access request, and an event outside the inspected coverage. These conditions should not share one negative label. Measure whether the system distinguishes them, acquires more evidence when appropriate, or abstains. Multiple-choice scoring alone can force a guess and conceal this distinction, so an auxiliary evidence-sufficiency label is useful even when the main benchmark requires an option.

11.7 Statistical interpretation and reproducibility

The diagnostic unit should follow the source dependence. Several questions about one video share appearance, events, and acquisition work; they are not interchangeable with the same number of questions drawn from independent videos. Confidence intervals and error analyses should respect that grouping. When a selector is stochastic, separate variability across random seeds from variability across source videos.

Predefine the operating budgets and comparison settings before inspecting test outcomes. Selecting a different budget for each method on the test set can produce an optimistic frontier. A validation set can guide allocation weights, resolution choices, and stopping thresholds, while a held-out test set assesses the resulting policy. Changes to prompts, image processors, or external tools should be recorded as changes to the system configuration.

Report the population used for every aggregate. Time-outs, malformed answers, unavailable sources, and tool failures can change the denominator. A complete-system report should retain these outcomes and identify their cause. Conditional accuracy on successfully processed examples can be informative as an additional statistic, but it should be accompanied by coverage and total completion rate.

Finally, a reproducible release should connect predictions to configuration and evidence traces. Checkpoint identifiers, software versions, preprocessing rules, source-access conventions, seeds, prompts, and output parsing are necessary to reconstruct the tested system. Trace fields should use stable units and identify generated intermediates separately from observations. This makes the proposed acquisition–representation–reasoning distinction testable without claiming that attention scores alone explain a model’s behavior.

12 Design Principles and Open Problems

12.1 Coverage versus concentration

A query about one fleeting fact favors concentrated search; an event-order query may require several separated regions. A single fixed scoring temperature or coverage weight cannot be expected to suit both. A promising direction is to infer evidence structure from the question while retaining exploration for uncertain intervals. This is a hypothesis to test with question-type ablations, not an established superiority claim.

The danger of pure relevance ranking is especially clear for relational questions. “What changed after the person returned?” requires a reference state and a later state; either frame alone can look relevant. A selector should be evaluated for evidence-set completeness, not just the average similarity of chosen frames.

12.2 Motion-preserving acquisition

Frames can identify objects, but actions often require temporal neighborhoods. Clip-level acquisition, local dense bursts, and motion-aware redundancy tests can preserve transitions at a modest cost. Evaluation should distinguish static detail, state change, counting, and action order. A selector optimized for still-frame matching may need different scoring features for these categories.

12.3 Uncertainty and stopping

An adaptive system must decide where to inspect next and when further inspection is no longer worth its cost. Selector uncertainty and answer uncertainty are different. A relevance model can be uncertain about a region even when the answerer is overconfident in a guessed option. Calibration should therefore be evaluated at both levels. Budget-aware stopping needs tests on distractors, repeated scenes, absent evidence, and questions that cannot be answered from the supplied video.

12.4 Memory with source recovery

The desirable memory is not necessarily the smallest summary. A system should retain provenance and enough structured state to recover exact details from original evidence. Candidate representations include event boundaries, entity tracks, timestamped summaries, and links to audio or OCR. Long-horizon identity consistency and correction of earlier memory errors are important evaluation targets.

12.5 Multimodal alignment

Audio, subtitles, OCR, and visual events can provide complementary clues. They can also conflict. Speech can describe an object before it appears; subtitles may lag; OCR can repeat across scenes. Retrieval should maintain modality-specific timestamps and uncertainty rather than concatenate all text indiscriminately. Tests should include missing audio, transcript errors, and asynchronous evidence, with separate budgets for each modality.

12.6 Joint temporal and spatial budgets

The allocation problem extends beyond the number of frames. One question needs a small written label; another needs many low-resolution views of a route. A principled scheduler can allocate token density, clip duration, and acquisition effort jointly. Fixed-token comparisons are crucial to establish whether improvements result from a better allocation or simply a larger effective input.

12.7 Training and deployment mismatch

Selection or compression trained with one answerer can exploit its particular representations and fail after the backbone changes. Robust studies should test cross-backbone transfer,

novel question types, and different temporal distributions. Reproducible tools and stable checkpoint identifiers make this more informative than comparisons against moving API baselines.

Production systems additionally face decode limits, storage access, throughput targets, and tail latency. A highly accurate iterative agent may be appropriate for offline archive search but unsuitable for a strict real-time service. Scheduling, caching, and graceful degradation therefore belong in an efficiency study, even when the core algorithm is model-agnostic.

13 Limitations and Responsible Use

This paper is a curated literature synthesis, not an exhaustive catalog or a meta-analysis. It includes no new experimental results. Numerical entries preserve the scope of the cited source and should not be treated as a standardized ranking. Some method internals and cost claims require full implementation-level reproduction beyond the source records used here. The taxonomy emphasizes QA and efficient evidence handling; video generation and task-specific recognition are not comprehensively reviewed.

Long-video models can expose private speech, personal behavior, and sensitive identities. Deployments should observe data permissions, minimize retained data, and limit access to source footage. Evidence traces can improve auditability while also creating additional sensitive records. A timestamped answer is more inspectable, but it does not guarantee that the inference is correct.

14 Conclusion

Efficient long-video understanding is best viewed as a chain of evidence decisions. Frame selection controls what is observed; compression controls what survives; memory controls what remains accessible; retrieval and agents control what can be revisited; context scaling controls how much the reasoner can process at once. These mechanisms are complementary, and their interaction is more informative than a frame-count-only comparison. The reviewed literature demonstrates several workable approaches, while leaving substantial questions about evidence completeness, recoverability, calibration, and full-system cost. A reproducible next step is to separate controlled selector and representation studies from complete-system evaluation, and to report grounded accuracy together with actual acquisition and serving budgets.

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